

Week Summary

Floating point arithmetic

Floating point arithmetic

- This week we have looked more closely at floating point numbers
- IEEE754 floating point representation of numbers:
 - Range and accuracy
 - Exceptions
- Helps us understand how a program has gone wrong
- Helps us quantify machine round-off vs numerical errors

Further floating point information

- If you would like to find out more about floating point arithmetic:
 - “Hypot – A story of a ‘simple’ function” <https://walkingrandomly.com/?p=6633>
 - “What every computer scientist should know about floating point arithmetic”, https://docs.oracle.com/cd/E19957-01/806-3568/ncg_goldberg.html

Peak performance

- We also saw how to calculate the theoretical maximum performance of a computer (R_{peak})
- This is the maximum rate at which floating point calculations could be performed if all the floating point hardware was used all the time
- In practice the real world performance can be much less than R_{peak} and over the next two weeks we will see a number of reasons why this is so